

DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY

Faculty of Engineering & Technology - Institute of Engineering & Technology

NAAC Accredited Grade "A++" | Established 1957 | Gorakhpur, UP, India

OFFICIAL CURRICULUM & DEGREE STRUCTURE - B.TECH MECHANICAL ENGINEERING

Four-Year Undergraduate Degree Program | Choice Based Credit System (CBCS) & NEP 2020

Effective Academic Sessions: 2024-25, 2025-26 | Department of Mechanical Engineering

1. PROGRAM EDUCATIONAL OBJECTIVES (PEOs) & VISION

The B.Tech in Mechanical Engineering program equips students with rigorous foundational principles in thermal science, fluid mechanics, solid mechanics, advanced manufacturing, CAD/CAM/CAE, robotics, and industrial automation to design sustainable solutions for modern industry.

2. TOTAL CREDIT DISTRIBUTION & DEGREE REQUIREMENTS

- Basic Sciences & Humanities (BSH): 28 Credits
- Engineering Science Courses (ESC): 24 Credits
- Professional Core Courses (PCC - ME): 70 Credits
- Professional Elective Courses (PEC): 18 Credits
- Open Elective Courses (OEC): 12 Credits
- Workshop Practice, Project Work & Industrial Internship: 16 Credits
- Mandatory Non-Credit Audit Courses (Environmental Science, Disaster Management): Pass

Total Degree Completion Requirement: 168 Credits

3. FIRST YEAR COMMON ENGINEERING FOUNDATIONS

Mathematics-I, Mathematics-II, Engineering Physics, Engineering Chemistry, Engineering Mechanics, Basic Electrical & Electronics, Engineering Graphics, CAD Modeling & Central Workshop Practices.

B.TECH MECHANICAL ENGINEERING - SEMESTER-WISE COURSE STRUCTURE

Curriculum Breakdown: Semesters 3 to 6 Core Specializations

SEMESTER 3 (Credits: 23)

1. BME-301: Engineering Thermodynamics & Vapor Cycles (3-1-0) - 4 Credits
2. BME-302: Strength of Materials & Mechanics of Solids (3-1-0) - 4 Credits
3. BME-303: Materials Science & Engineering Metallurgy (3-0-0) - 3 Credits
4. BME-304: Fluid Mechanics & Hydraulic Machinery (3-1-0) - 4 Credits
5. BHS-301: Technical Writing & Universal Professional Ethics (2-0-0) - 2 Credits
6. BME-351: Strength of Materials & Hardness Testing Lab (0-0-3) - 1.5 Credits
7. BME-352: Fluid Mechanics & Flow Measurement Lab (0-0-3) - 1.5 Credits
8. BME-353: Machine Drawing & SolidWorks 3D CAD Lab (0-0-4) - 2 Credits

SEMESTER 4 (Credits: 23)

1. BME-401: Applied Thermodynamics & IC Engines (3-1-0) - 4 Credits
2. BME-402: Kinematics of Machines & Mechanism Synthesis (3-1-0) - 4 Credits
3. BME-403: Manufacturing Processes & Machine Tools (3-1-0) - 4 Credits
4. BME-404: Measurement, Metrology & Instrumentation (3-0-0) - 3 Credits
5. BAS-401: Advanced Mathematics & Numerical Methods (3-0-0) - 3 Credits
6. BME-451: IC Engines, Fuels & Lubricants Lab (0-0-3) - 1.5 Credits
7. BME-452: Manufacturing Technology & Machine Shop Lab (0-0-3) - 1.5 Credits

SEMESTER 5 (Credits: 22)

1. BME-501: Dynamics of Machines & Mechanical Vibrations (3-1-0) - 4 Credits
2. BME-502: Heat & Mass Transfer Engineering (3-1-0) - 4 Credits
3. BME-503: Design of Machine Elements - I (3-1-0) - 4 Credits
4. BME-504: Industrial Engineering & Operations Research (3-0-0) - 3 Credits
5. Professional Elective I: Non-Conventional Energy Resources (3-0-0) - 3 Credits
6. BME-551: Heat Transfer & Thermal Conductivity Lab (0-0-3) - 1.5 Credits
7. BME-552: Dynamics & Balancing Lab (0-0-3) - 1.5 Credits

SEMESTER 6 (Credits: 22)

1. BME-601: Design of Machine Elements - II (Gears, Bearings) (3-1-0) - 4 Credits
2. BME-602: Refrigeration and Air Conditioning (RAC) (3-1-0) - 4 Credits
3. BME-603: Fluid Power Control & Mechatronics (3-0-0) - 3 Credits
4. Professional Elective II: Automobile Engineering & EV Technology (3-0-0) - 3 Credits
5. Open Elective I: Project Management & Engineering Economics (3-0-0) - 3 Credits
6. BME-651: Refrigeration & Air Conditioning Lab (0-0-3) - 1.5 Credits
7. BME-652: Mini Project & Industrial Summer Training Evaluation - 2 Credits

FINAL YEAR CURRICULUM & DEGREE ORDINANCES - B.TECH MECHANICAL

Semesters 7 & 8: Advanced Automation, Capstone & Examination Policy

SEMESTER 7 (Credits: 21)

1. BME-701: Computer Aided Design & Manufacturing (CAD/CAM/CNC) (3-1-0) - 4 Credits
2. BME-702: Power Plant Engineering & Gas Turbines (3-0-0) - 3 Credits
3. Professional Elective III: Industrial Robotics & Automation (3-0-0) - 3 Credits
4. Open Elective II: Total Quality Management (TQM) & Six Sigma (3-0-0) - 3 Credits
5. BME-751: CAD/CAM, CNC Machining & Simulation Lab (0-0-3) - 1.5 Credits
6. BME-752: 6-Week Industrial Internship / Industrial Training Evaluation - 2.5 Credits
7. BME-753: Capstone Major Project - Phase I (Fabrication Planning & Design) - 4 Credits

SEMESTER 8 (Credits: 18)

1. BME-801: Finite Element Method & Computational Fluid Dynamics (CFD) (3-1-0) - 4 Credits
2. Professional Elective IV: Advanced Welding & Additive Manufacturing (3D Printing) (3-0-0) - 3 Credits
3. Professional Elective V: Supply Chain Management & Logistics (3-0-0) - 3 Credits
4. BME-851: Capstone Major Project - Phase II (Fabrication, Testing & Thesis Defense) - 6 Credits
5. BME-852: Comprehensive Technical Viva-Voce & Grand Viva - 2 Credits

EXAMINATION & CONTINUOUS EVALUATION SCHEME

Theory Courses: 100 Marks (30 Marks Mid-Term Assessment + 70 Marks University End-Term Exam).

Practical Labs: 100 Marks (50 Marks Continuous Workshop/Lab + 50 Marks University External Viva).

Passing Standards: 40% in theory papers, 50% in laboratory examinations, minimum CGPA 5.0 for award.

OFFICIAL SIGNATURES & VERIFICATION STAMP:

Prof. Dean, Faculty of Engineering & Technology
Deen Dayal Upadhyaya Gorakhpur University, UP

Controller of Examinations
DDU Gorakhpur University